



CRYPTOGRAPHY AND NETWORK SECURITY

Goal to learn

- ❑ Computer Security, Network Security, Internet Security
- ❑ Security Goal
- ❑ Security Attack
- ❑ Passive Attack & Active Attack
- ❑ Cryptography – Overview
- ❑ Basic Terminology
- ❑ Public Key – Private Key Cryptography
- ❑ Types of Encryption Operations



Network Security

Computer Security - generic name for the collection of tools designed to protect data and to thwart hackers.

Network Security - measures to protect data during their transmission.

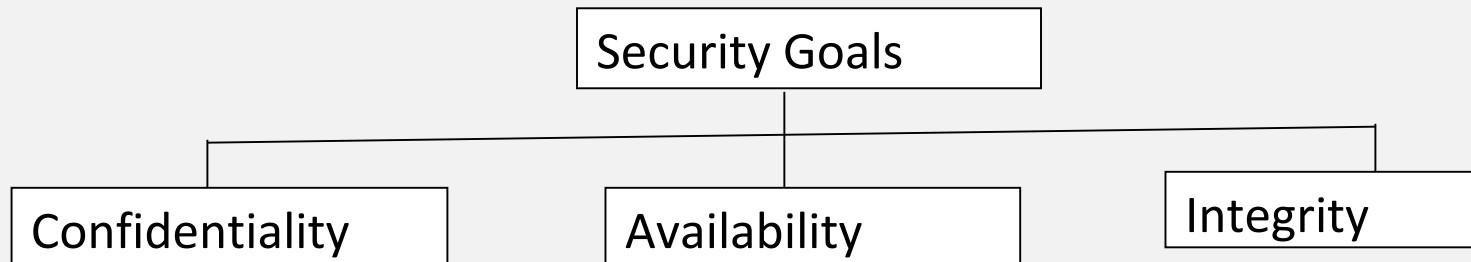
Internet Security - measures to protect data during their transmission over a collection of interconnected networks.



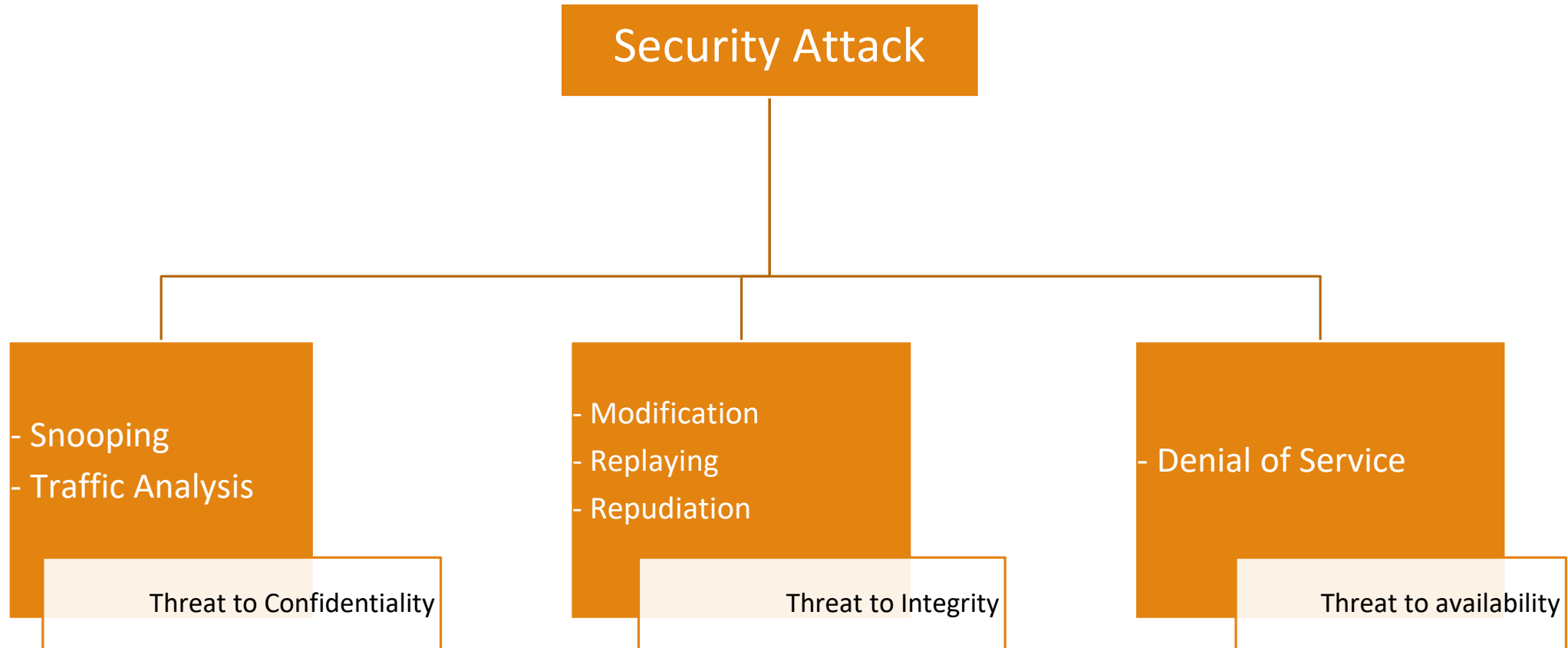
Security Goals

Information needs to be hidden from -

- ❑ Unauthorized access (Confidentiality)
- ❑ Protected from unauthorized change (integrity)
- ❑ Available to an authorized entity when it is needed (availability)



Security Attack



Attacks Threatening Confidentiality

Snooping -

Unauthorized access to or interception of data.

Traffic analysis -

Can find some other type information by monitoring online traffic.

He can find the electronic address (such as e-mail address) of the sender or the receiver.



Attacks Threatening Integrity

- **Modification** - After intercepting or accessing information, the attacker modifies the information to make it beneficial to himself
- **Masquerading**- Masquerading or spoofing(trick) happens when the attacker impersonates somebody else.
- **Replaying**- The attacker obtains a copy of a message sent by a user and later tries to replay it.
- **Repudiation**- Performed by one of the two parties in the communication: the sender or the receiver.
 - The sender of the message might later deny that she has sent the message.
 - The receiver of the message might later deny that he has received the message.

Attacks Threatening Availability

- It may slow down or totally interrupt the service of a system.
- He might send so many bogus requests to a server that the server that the server crashes because of the heavy load.
- The attacker might intercept and delete a server's response to a client, making the client to believe that the server is not responding.

Passive Attacks – Active Attack

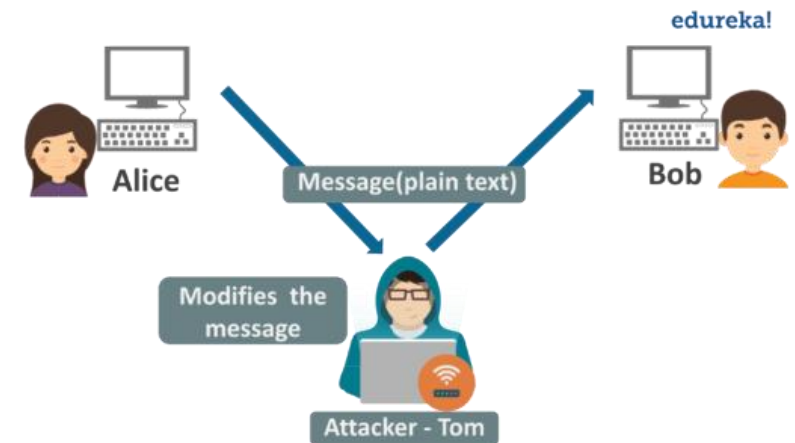
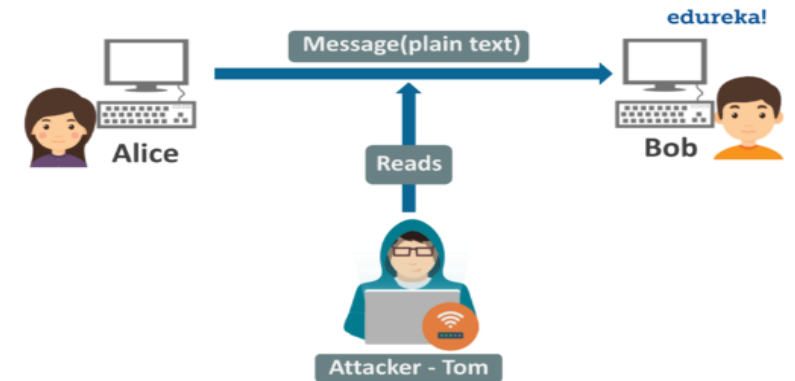
A passive attack is a network attack in which a system is ***monitored*** and sometimes ***scanned for open ports and vulnerabilities***.

The purpose is solely to gain ***information about the target*** and ***no data is changed on the target***.

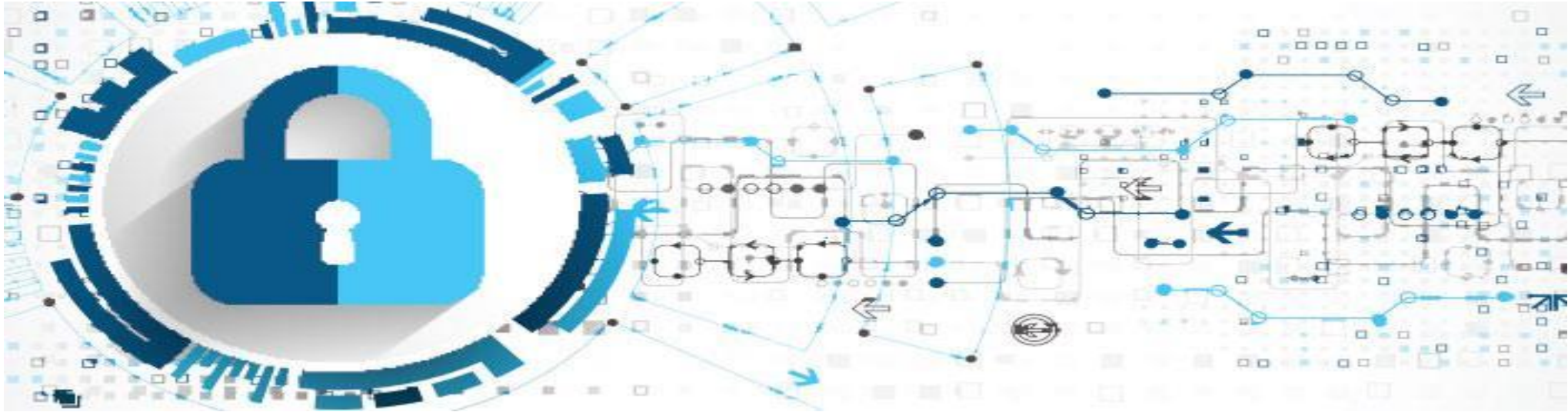
An active attack attempts to ***alter system resources*** or ***affect their operations***. Active attack involve some ***modification*** of the data stream or ***creation of false statement***.

Types of active attacks:

masquerade, session replay, denial of service (DoS) attack.



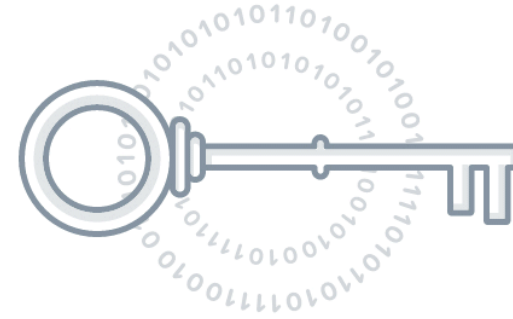
Cryptography



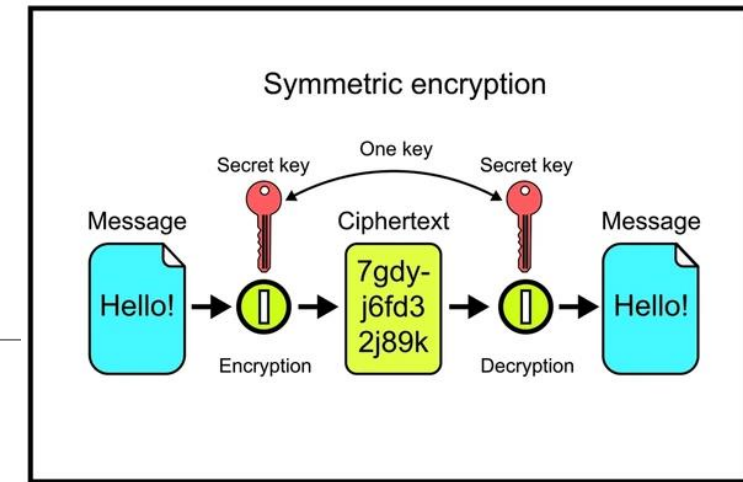
Cryptography is the **practice of concealing and unmasking information** using **coded algorithms, hashes, and signatures** to **protect it from unauthorized access**.

Cryptography

- ✓ Cryptography is the study of
 - Secret (crypto-) writing (-graphy)
- ✓ Concerned with developing algorithms.
- ✓ Verify the correctness of a message to the recipient (authentication)
- ✓ Form the basis of many technological solutions to computer and communications security problems



Some Basic Terminology



- ❑ **Plaintext** - Original message

- ❑ **Ciphertext** - Coded message.

[Ciphertext is an encrypted message that is created from a plaintext message using encryption techniques]

- ❑ **Cipher** [encryption techniques] - Algorithm for transforming plaintext to ciphertext

- ❑ **Secret Key** - Info used in cipher known only to sender/receiver.

[Secret key is a value/string/text file used by cipher to encode and decode the plain text to cipher text and vice versa.]

- ❑ **Encipher (encrypt)** - Converting plaintext to ciphertext

- ❑ **Decipher (decrypt)** - recovering plaintext from ciphertext

- ❑ **Cryptography** - study of encryption principles/methods.

- ❑ **Cryptanalysis (codebreaking)** - study of principles/ methods of deciphering cipher-text *without* knowing key.

TYPES OF CRYPTOGRAPHY

1 Secret Key
Cryptography (SKC)

2 Public Key
Cryptography (PKC)

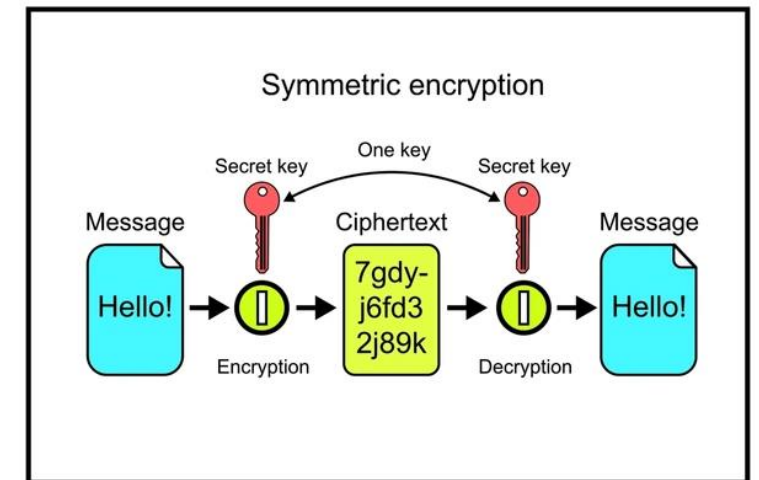
3 Hash Functions

1: Symmetric Encryption

Symmetric encryption is a method of encryption that uses the same key to both encrypt and decrypt data. It's often used for large data sets and high-volume data processing because it's fast and efficient.

How it works

A single key is used to encrypt and decrypt data. For example, if you encrypt an email with a key and send it to a friend, they can use the same key to decrypt it.



Strengths: Symmetric encryption is simple and efficient. It's also strong cryptographically per bit of key.

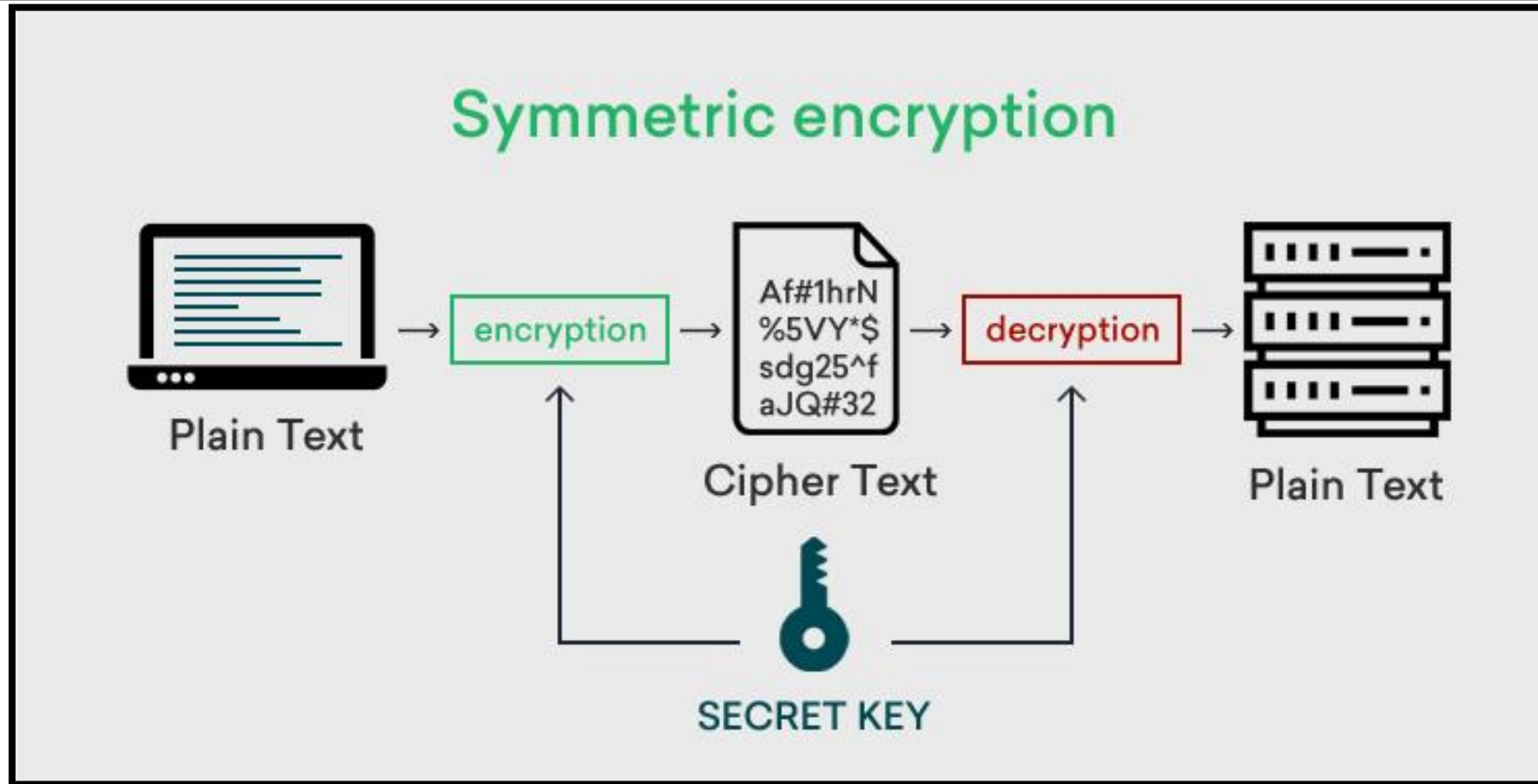
Weaknesses: Symmetric encryption can be less secure than asymmetric encryption because it relies on secure key exchange. If the key is intercepted, the data can be accessed.

Real-world use

Symmetric encryption is often used in combination with asymmetric encryption in secure communication protocols. Asymmetric encryption uses different keys for encryption and decryption, which can help with secure key distribution.

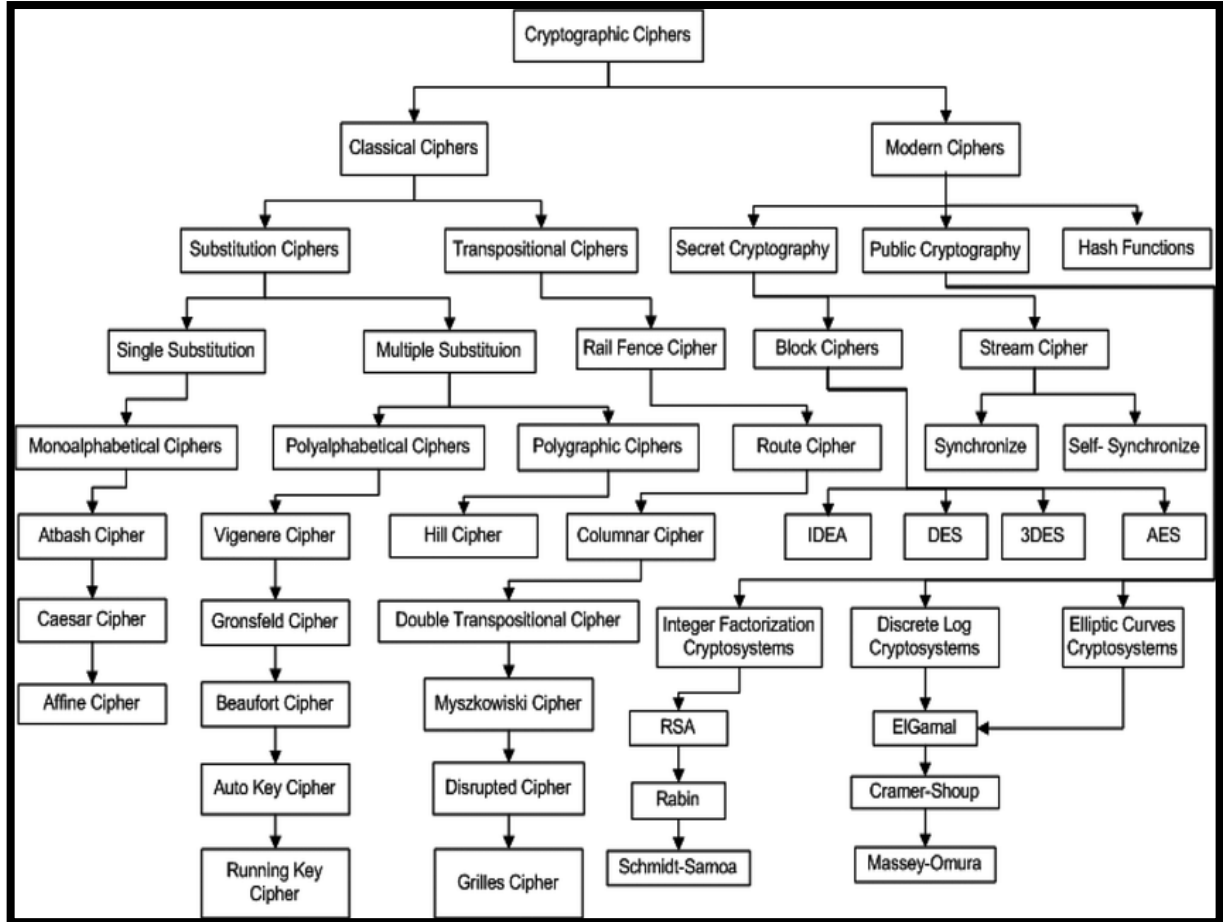
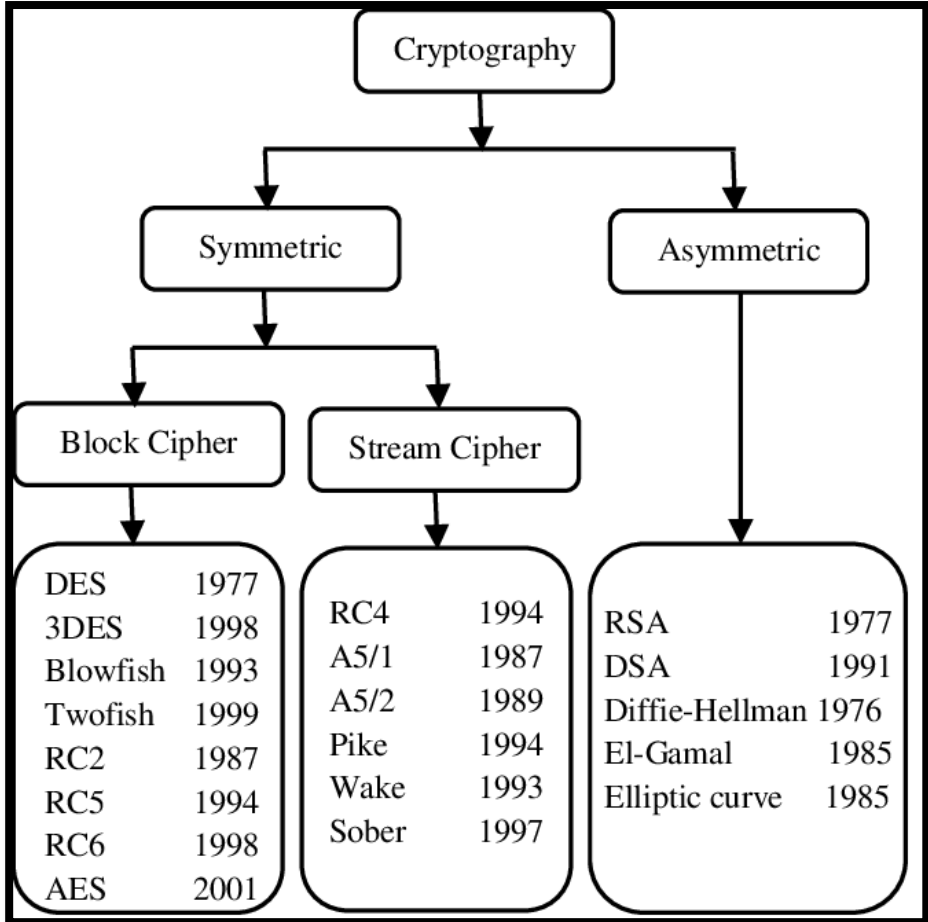
- **Examples:** The Caesar Cipher and the Vigenère Cipher are both examples of symmetric encryption techniques.

Private-Key / Symmetric Encryption



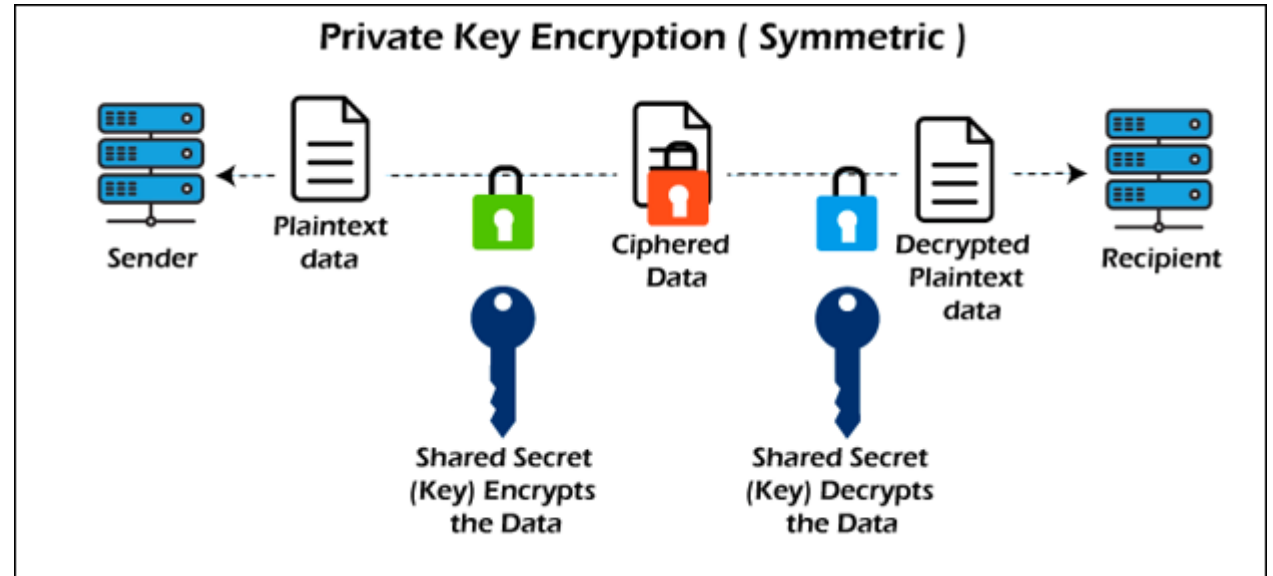
Examples of Cipher

<https://www.devglan.com/online-tools/text-encryption-decryption>



Way to securely share the secret key!

- ❑ Pre-shared key
- ❑ Key exchange protocols
- ❑ Key distribution centers
- ❑ Public key encryption (Hybrid)



The Number of keys required for N number of clients= $N(N-1)/2$

The disadvantages of symmetric keys include:

Key Distribution: Securely sharing the same key is challenging and risky.

Scalability: Requires a unique key for each pair of users, making management difficult as networks grow.

No Non-Repudiation: Both parties use the same key, so it's unclear who sent or received a message.

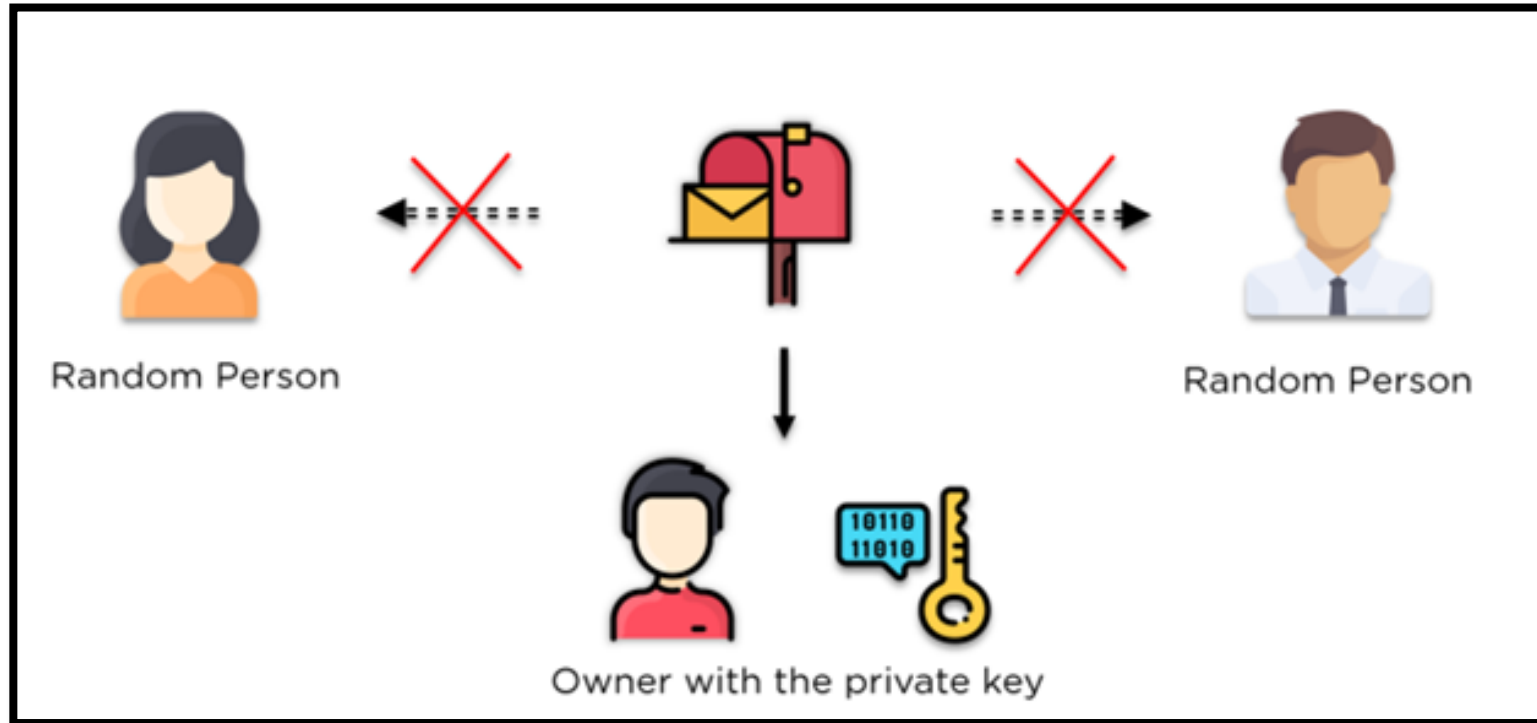
Key Compromise: If the key is exposed, all past and future encrypted data is at risk.

No Forward Secrecy: Compromised keys reveal all past communications, unlike some asymmetric methods.

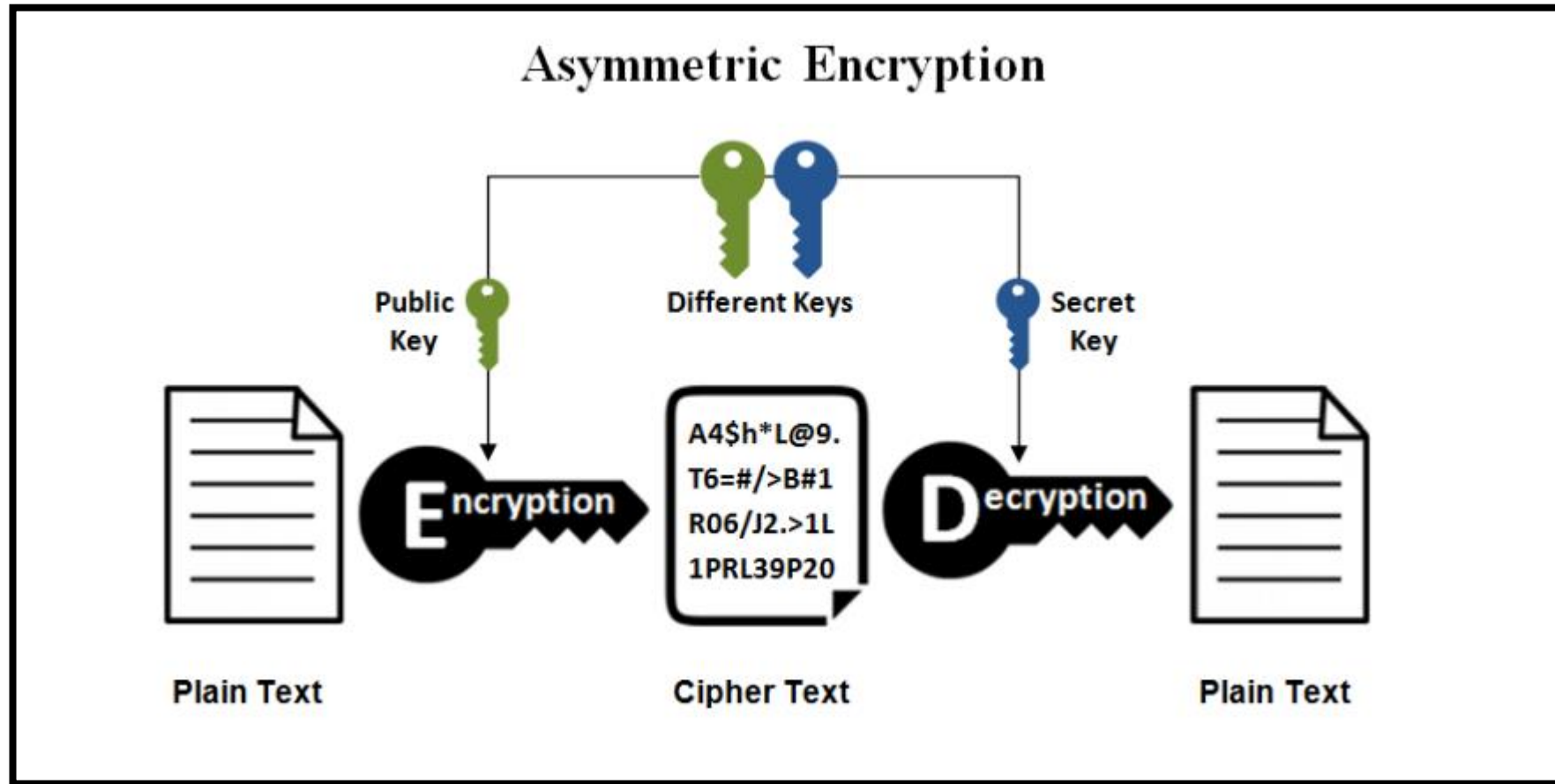
2: Asymmetric Encryption

Asymmetric encryption, also known as public key cryptography, is a method of encrypting and decrypting data using a **public and private key pair**:

- **Encryption:** The public key is used to encrypt plaintext into ciphertext.
- **Decryption:** The corresponding private key is used to decrypt the ciphertext



Public-Key / Asymmetric Cryptography





Private key



Public key



Bob's public key



Private key



Public key



Alice's public key



Encrypted with
Bob's public key



Decrypted with
Bob's private key

Public Key Cryptography / Asymmetric Cryptography

- Type of encryption that operates using a public and private key pair.
 - Public key may be disseminated widely.
 - Security is reliant of the private key being known only to the owner.
- The public key is used to encrypt a message. Only the corresponding private key can decrypt that message.
- Keys are simply numbers. Numbers with more digits are more secure.

The Number of keys required for N number of clients= $2N$

Where is Asymmetric Key Cryptography used?



Digital Signatures to maintain authenticity of documents



Encrypted browsing sessions for better protection against hackers



Managing Crypto-currency transactions securely



Sharing Keys for Symmetric Key Cryptography

Uses of Asymmetric/Public Key cryptography

Asymmetric cryptography, or public-key cryptography, uses two keys (public and private) and is essential in:

- 1.Secure Communication:** Encrypting data with a public key, only the corresponding private key can decrypt it.
- 2.Digital Signatures:** Verifying the sender's identity and message integrity through signatures signed with a private key.
- 3.Key Exchange:** Securely exchanging symmetric keys, used in protocols like Diffie-Hellman.
- 4.SSL/TLS:** Enabling secure web communication over HTTPS.
- 5.Digital Certificates:** Verifying identities through certificates issued by trusted authorities.
- 6.Email Security:** Encrypting emails (e.g., with PGP) for privacy.
- 7.Cryptocurrency:** Protecting wallets and authenticating transactions.
- 8.Authentication:** Used in login systems, such as SSH, to verify user identity

Asymmetric cryptography offers several advantages that makes it secure is:

- 1. Enhanced Security:** Since public and private keys are different, the private key never needs to be shared, reducing the risk of unauthorized access.
- 2. Authentication:** Digital signatures authenticate the sender's identity, ensuring data integrity and preventing tampering.
- 3. Scalability:** Public keys can be distributed openly, making it easy to establish secure communication with multiple users without needing a pre-shared secret.
- 4. Non-Repudiation:** Digital signatures provide proof that a message was sent by a specific individual, preventing the sender from denying it.
- 5. Secure Key Exchange:** It enables secure exchange of symmetric encryption keys over an insecure channel, which is foundational for many secure communication protocols.

Disadvantages

Performance: Asymmetric encryption algorithms are typically slower than symmetric encryption algorithms, which can make them unsuitable for certain applications.

Key management: Asymmetric encryption requires the management of two keys for each user, which can be more complex than managing a single key in symmetric encryption.

Trust issues: Asymmetric encryption requires trust in the public key infrastructure, as there is a risk of false or malicious public keys being used.

Compatibility: Asymmetric encryption algorithms may not be compatible with all systems and applications, which can make it difficult to implement in certain environments.

More Encryption Operations -

Hashing

- Hash Function: A one-way function that converts any form of data into a unique sequence of bytes.
 - Input may be any data and have any size
 - The output hash is always the same size, regardless of input
 - One-Way meaning you cannot determine the input from the output
 - The same input always produces the same output
 - Changing even 1 bit of the input data will produce a completely different output